COMMODITY ANALYST

Hormuz Disruption May Impact Global Agriculture Prices

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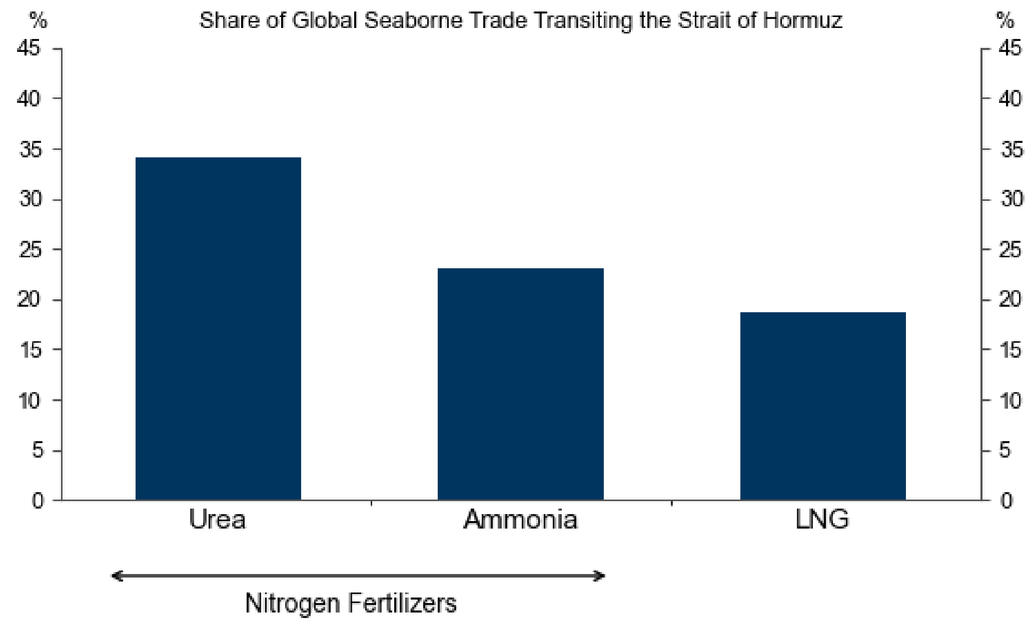
- The Strait of Hormuz is a critical route in the global nitrogen fertilizer market, which accounts for 60% of global fertilizer use and is especially important for crops like corn and other grains. More than a quarter of global nitrogen fertilizer trade and about 20% of LNG — the primary feedstock for nitrogen production — typically transit the Strait, meaning Hormuz disruptions both constrain global fertilizer availability and raise fertilizer production costs elsewhere.

- While higher fertilizer prices may eventually feed into grain prices — given fertilizer accounts for ~20% of grain costs — the largest potential boost to grain prices is more likely to come from reduced grain supply. Fertilizer disruptions may reduce grain production both through yield losses from delayed/sub-optimal nitrogen application, and potential acreage shifts toward less fertilizer-intensive crops. While the US is currently relatively insulated—because the conflict began just ahead of the planting season, by which point most farmers likely already secured fertilizers—more exposed regions such as Europe, Australia and the Southern Hemisphere, where crop calendars are later, could increase demand for US grain exports if fertilizer disruptions persist, raising US grain prices too.

- More broadly, the Middle East conflict underscores the role of commodities as a hedge against supply disruptions. Roughly 80% of the BCOM basket is either directly or indirectly exposed to the conflict via supply losses, a configuration that is likely to raise inflation while weighing on growth—an unfavorable mix for both bonds and equities. With supply increasingly concentrated across commodity markets and the growing use of commodities as tools of leverage—consistent with our commodity control cycle framework—we continue to emphasize the insurance value that commodities can provide within portfolios.

More Than a Quarter of Global Nitrogen Fertilizer Trade and ~ 20% of LNG — the Primary Feedstock for Nitrogen Production — Typically Transit the Strait of Hormuz



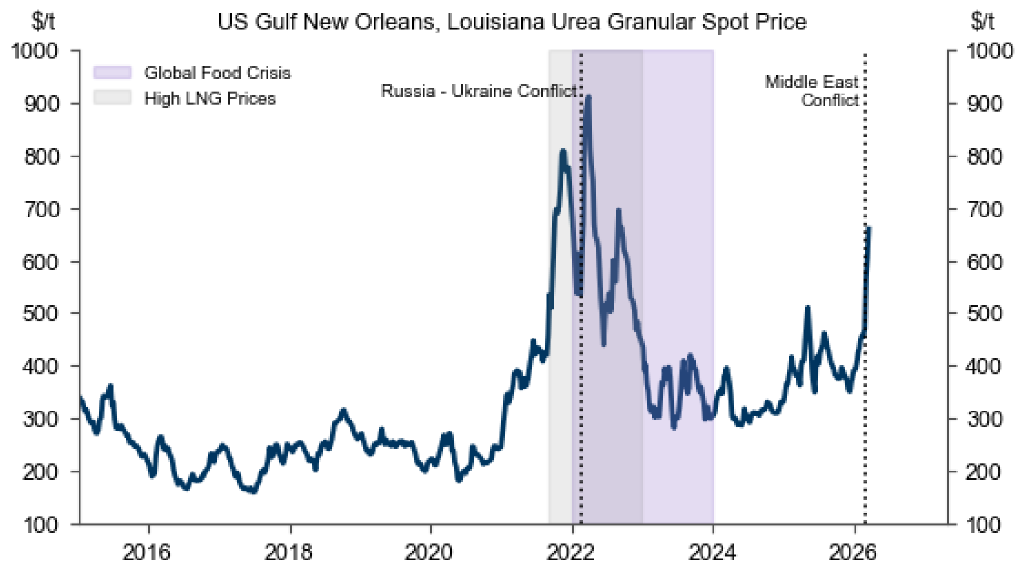
We show annual averages for 2025 for LNG and for 2024 for urea and ammonia.

Source: IFA, CRU, Goldman Sachs Global Investment Research

Hormuz Disruption May Impact Global Agriculture Prices[1]

The Strait of Hormuz is a critical route in the global nitrogen fertilizer market, which accounts for 60% of global fertilizer use and is especially important for crops like corn and other grains. Since the onset of the conflict, nitrogen fertilizer prices have risen by 40% (Exhibit 1). In this *Commodity Analyst*, we discuss the implications for agricultural markets.

Exhibit 1: Since the Onset of the Middle East Conflict, Nitrogen Fertilizer Prices Like Urea Have Risen by 40%



Source: Bloomberg, Goldman Sachs Global Investment Research

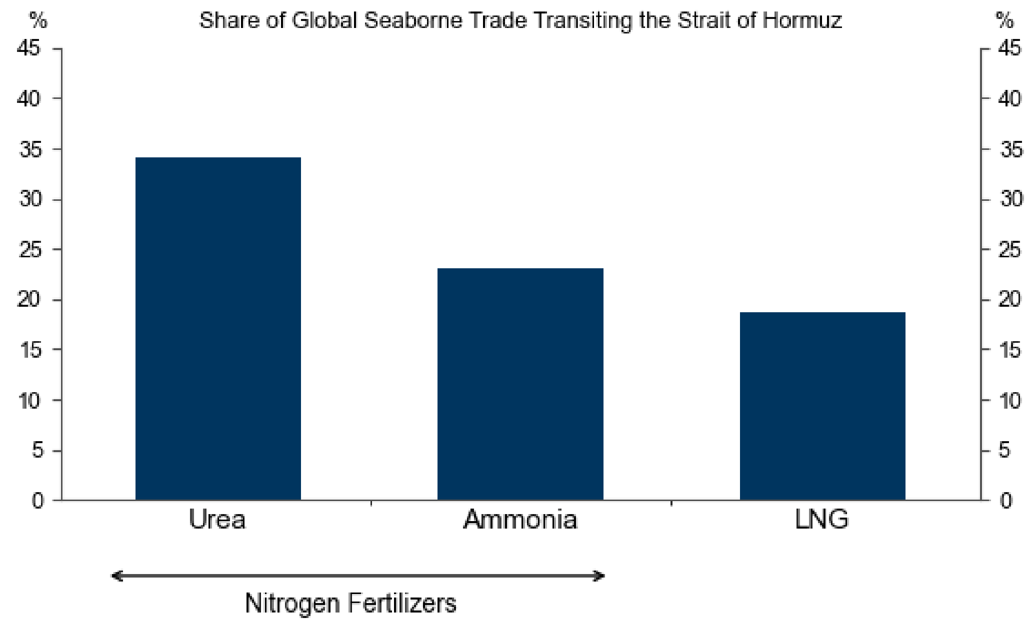
Hormuz Disruptions Constrain Global Fertilizer Availability and Raise Fertilizer Production Costs Elsewhere

More than a quarter of global nitrogen fertilizer trade and about 20% of LNG — the primary feedstock for nitrogen production — typically transit the Strait, meaning Hormuz disruptions both constrain global fertilizer availability[2] and raise fertilizer production costs elsewhere (Exhibit 2).

[1] The authors would like to thank Ishan Kalia –an intern on our commodities research team– for his extensive contributions.

[2] Even if shipping flows resume, fertilizer supply may remain constrained. QatarEnergy — responsible for roughly 10% of global urea trade — halted urea and ammonia production at its Ras Laffan complex following a recent drone attack, while Iran has reportedly taken all ammonia production offline.

Exhibit 2: More Than a Quarter of Global Nitrogen Fertilizer Trade and ~ 20% of LNG — the Primary Feedstock for Nitrogen Production — Transit the Strait



We show annual averages for 2025 for LNG and for 2024 for urea and ammonia.

Source: IFA, CRU, Goldman Sachs Global Investment Research

Spare fertilizer production capacity outside the Middle East appears limited: Russia—typically accounting for around 15% of global nitrogen fertilizer exports—is constrained by recent Ukrainian attacks on production facilities and domestic export limits aimed at prioritizing local needs, while China's fertilizer export restrictions, set to expire in August, are increasingly likely to be extended amid recent market shocks.

The Impact of Higher Fertilizer Prices On Grains Prices

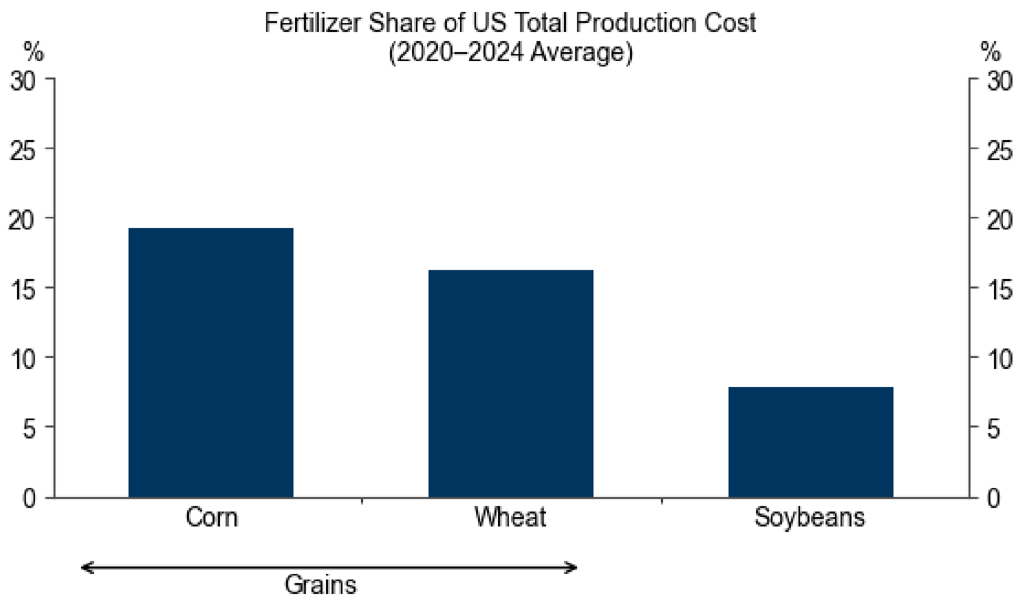
While higher fertilizer prices may eventually feed into grain prices — given fertilizer accounts for ~20% of grain costs — the largest potential boost to grain prices could come from reduced grain supply.[3]

Fertilizer disruptions may reduce grain production both through yield losses from delayed/sub-optimal nitrogen application,[4] and potential acreage shifts toward less fertilizer-intensive crops like soybeans (Exhibit 3).

[3] Based on cost pass-through analysis alone, our US economists estimate that higher fertilizer costs could lift food prices by around 1.5% this year, adding roughly 0.1pp to headline inflation.

[4] In the US, roughly 50% of nitrogen is applied pre-planting (March–April) and a further 25% post-planting (May–June) for corn.

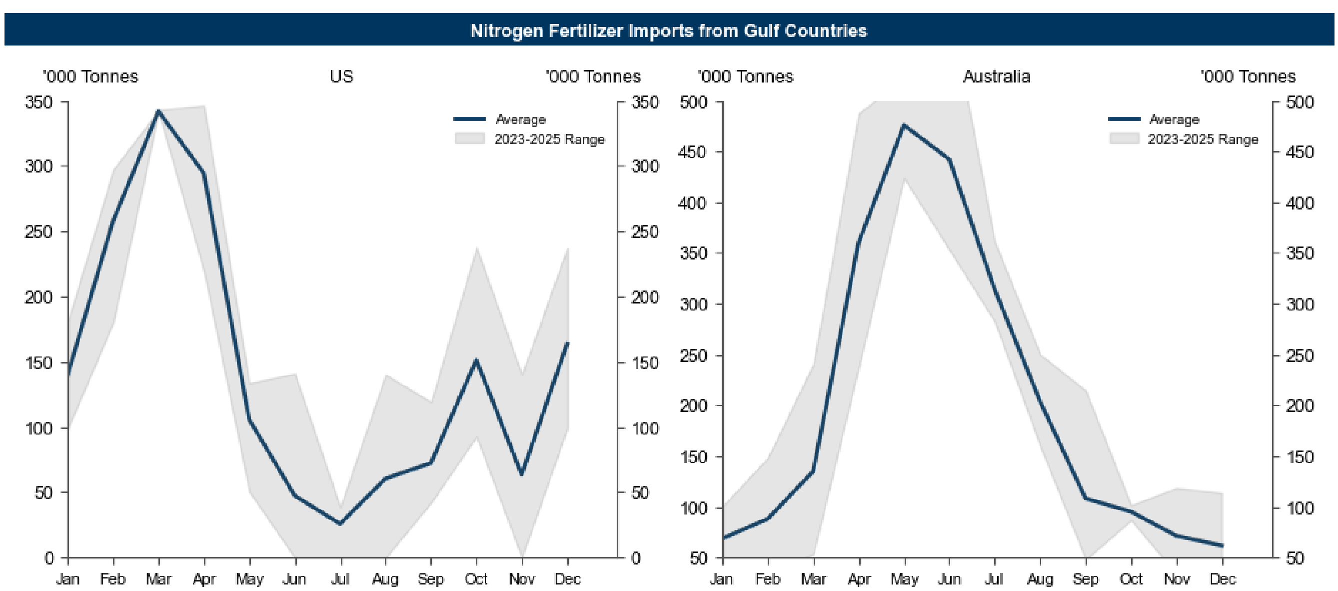
Exhibit 3: Some Farmers May Shift Acreage to Less Fertilizer-Intensive Crops Like Soybeans as Fertilizer Prices Rise



Source: USDA, Goldman Sachs Global Investment Research

The US is initially likely relatively insulated from the fertilizer disruption, because the conflict began just ahead of the planting season, by which point most farmers likely already secured fertilizers. Yet, because US fertilizer imports peak in February-April and Gulf-to-US shipments take roughly a month, disruptions to March loadings could still affect April availability (Exhibit 4). With no strategic reserves and limited ability to rapidly scale domestic fertilizer production — despite access to US natural gas —, the US may retain some exposure to reduced global fertilizer supply during peak demand season.

Exhibit 4: Because of Shipping Lags, March Disruptions in the Gulf Could Impact April US fertilizer Availability; Extended Disruptions May Pose Significant Risks for Import-Dependent Australia



Source: Global Trade Tracker, Goldman Sachs Global Investment Research

More exposed regions such as Europe, Australia and the Southern Hemisphere, where crop calendars are later and domestic fertilizer production is vulnerable to LNG price volatility, could increase demand for US grain exports if fertilizer disruptions persist, putting upward pressure on US prices.

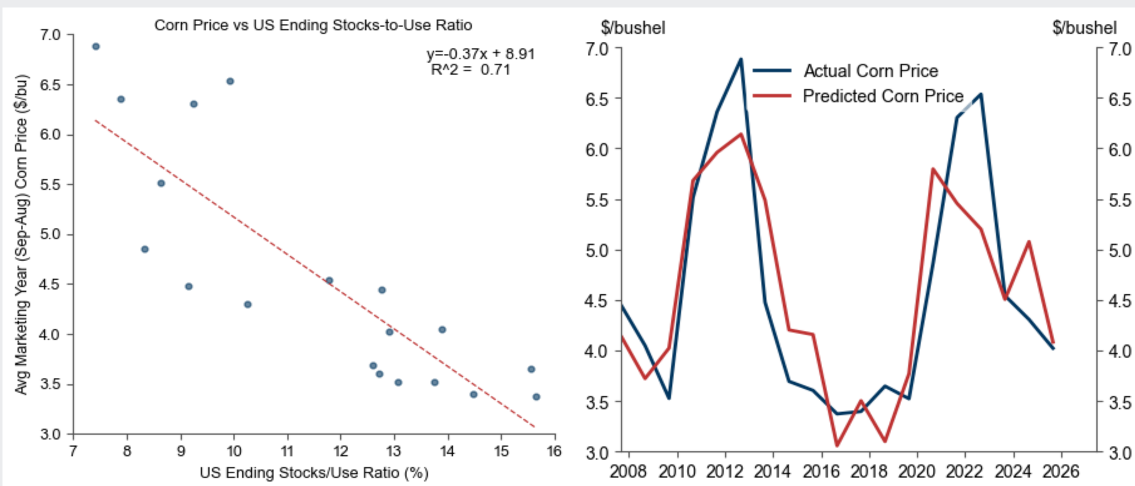
A Simple US Corn Pricing Model

We formalize the hypothesis that global fertilizer supply disruptions may reduce grain production and raise US grain prices using a simple US corn pricing framework.

Corn is produced once per year but consumed continuously. As a result, the marketing-year price (September-August) must clear the annual supply-demand balance — high enough to avoid running out of inventory before the next harvest, but low enough to prevent excessive leftovers. Prices, therefore, move inversely with the "leftover stocks" or "ending stocks"-to-use ratio (Exhibit 5).

Ending stocks equal **beginning stocks** carried over from the prior harvest, plus **production**, minus **total use** (domestic consumption and exports). Production itself depends on **planted acreage** and **yield per acre**.

Exhibit 5: Corn Prices Move Inversely With the Ending Stocks-to-Use Ratio



Source: USDA, Goldman Sachs Global Investment Research

Fertilizer disruptions may raise corn prices by lowering the ending stocks-to-use ratio through three channels:

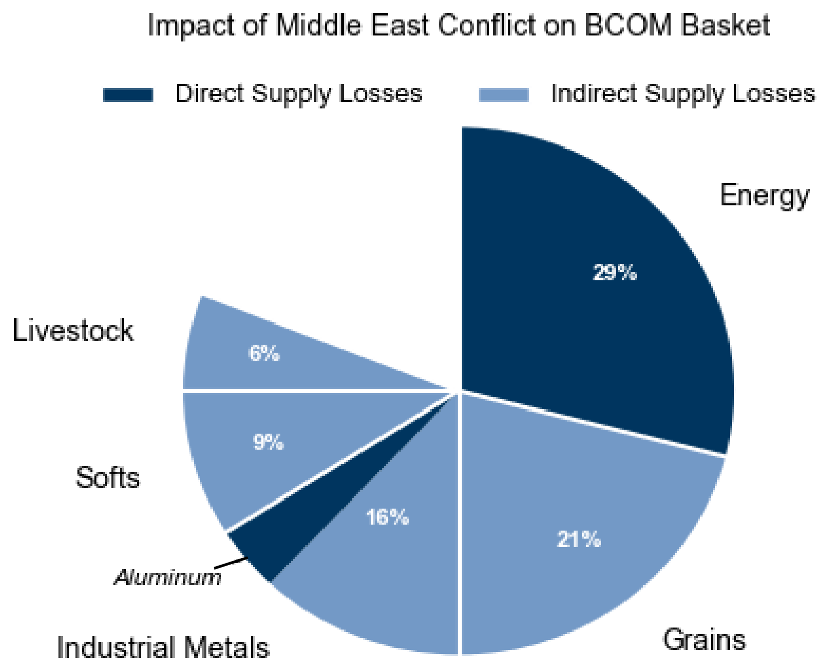
1. **Lower yields** due to delayed or sub-optimal nitrogen application (with outcomes contingent on weather conditions);
2. **Reduced acreage**, as some farmers shift toward less fertilizer-intensive crops such as soybeans; and
3. **Higher exports**, as more fertilizer-exposed regions — such as Europe and potentially the Southern Hemisphere—increase demand for US corn.

The Insurance Value of Commodities In Portfolios

More broadly, the Middle East conflict underscores the role of commodities as a hedge against supply disruptions. Roughly 80% of the BCOM basket is either directly or indirectly exposed to the conflict via supply losses, a configuration that is likely to raise inflation while weighing on growth—an unfavorable mix for both bonds and equities (Exhibit 6).

The Gulf region is a major producer of energy and aluminum, and also supplies key chemical inputs such as sulfur—critical for the production of industrial metals like copper—and nitrogen, which underpins agricultural production (softs and grains). Agricultural commodities, in turn—particularly grains—serve as a key feedstock for livestock. Year-to-date, the total return for the BCOM basket stands at 20% (vs. 15% in 2025).

Exhibit 6: Roughly 80% of the BCOM Basket Is Either Directly or Indirectly Exposed to the Conflict via Supply Losses



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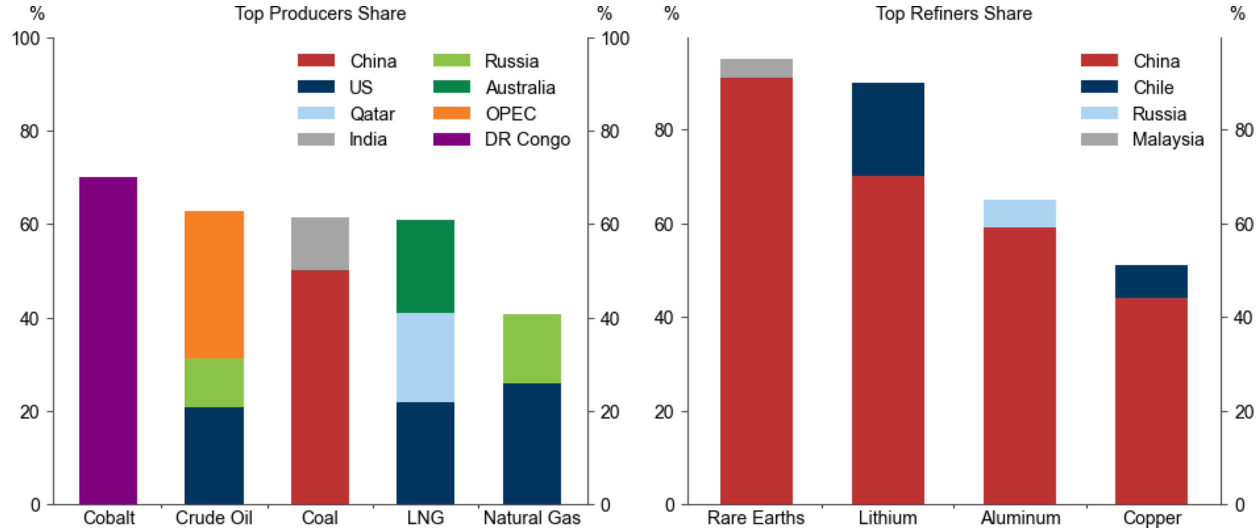
Source: Bloomberg, Goldman Sachs Global Investment Research

With supply increasingly concentrated across commodity markets and the growing use of commodities as tools of leverage—consistent with our Commodity Control Cycle framework—we continue to emphasize the insurance value that commodities can provide within portfolios (Exhibit 7).

Exhibit 7: Commodity Supply Is Increasingly Concentrated



Source: IEA, EIA, National Mining Association, Goldman Sachs Global Investment Research

Disclosure Appendix

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